

FOREST CITY, IA, USA

WMO: 720643

Lat: 43.235N

Lon: 93.624W

Elev: 375

StdP: 96.91

Time Zone: -6.00 (NAC)

Period: 09-19

WBAN: 54940

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.0	-21.1	-27.4	0.3	-23.3	-25.0	0.4	-21.0	12.9	-7.5	12.0	-7.8	2.6	330	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.3	21.6	30.9	21.4	28.9	20.9	24.8	28.8	23.7	27.9	22.8	26.7	5.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.0	18.6	26.7	22.5	18.0	26.0	22.0	17.4	25.3	77.3	29.2	72.9	27.6	69.3	27.1	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	10.1	8.9	-29.1	35.2	3.0	2.1	-31.3	36.7	-33.1	38.0	-34.8	39.2	-37.0	40.8		
(5)			-29.3	26.7	3.0	1.2	-31.5	27.5	-33.2	28.2	-34.9	28.8	-37.1	29.7	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-8.6	-6.9	1.1	7.8	14.7	20.5	22.0	20.3	17.0	9.0	1.6	-5.1	(6)	
	DBStd	12.00	7.30	6.52	6.99	5.35	5.10	3.34	3.19	3.20	4.39	4.88	5.67	6.13	(7)	
	HDD10.0	2256	576	474	286	102	14	0	0	0	3	79	256	467	(8)	
	HDD18.3	4158	834	707	534	316	134	15	7	15	76	292	502	725	(9)	
	CDD10.0	1481	0	0	10	37	162	316	373	321	213	47	3	0	(10)	
	CDD18.3	342	0	0	1	1	23	81	121	78	36	2	0	0	(11)	
	CDH23.3	3242	0	0	4	17	281	745	1133	691	337	34	0	0	(12)	
	CDH26.7	987	0	0	0	2	99	239	366	183	91	6	0	0	(13)	
(14) Wind	WSAvg	3.8	4.4	4.3	4.5	4.8	4.1	3.5	2.7	2.5	3.1	3.8	4.3	4.0	(14)	
(15) Precipitation	PrecAvg	774	18	19	45	77	100	115	110	100	79	53	35	24	(15)	
	PrecMax	1051	49	93	97	180	232	312	312	356	245	126	125	72	(16)	
	PrecMin	423	0	0	0	20	22	24	14	16	9	2	0	0	(17)	
	PrecStd	166	14	19	26	39	48	64	66	70	56	34	27	18	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	12.6	22.9	26.0	33.0	34.7	34.0	32.4	32.1	27.2	20.0	12.1	(19)	
		MCWB	4.5	10.3	14.7	14.3	19.7	21.7	23.9	21.2	20.2	15.5	12.6	8.3	(20)	
	2%	DB	3.0	6.1	17.9	22.4	29.8	31.4	32.3	30.3	29.8	22.9	15.9	7.2	(21)	
		MCWB	1.9	4.0	11.9	13.7	18.6	20.3	22.9	22.0	21.0	14.2	10.8	5.4	(22)	
	5%	DB	2.2	2.9	14.9	20.0	26.3	29.8	30.2	28.7	27.2	20.3	12.5	4.8	(23)	
		MCWB	1.3	1.8	10.4	12.1	16.3	19.6	22.7	21.9	19.8	13.9	8.4	3.0	(24)	
	10%	DB	0.9	1.4	10.9	17.3	23.0	27.4	28.0	27.3	25.1	17.4	10.0	2.4	(25)	
		MCWB	0.5	0.6	8.2	10.9	15.0	19.2	21.5	20.8	18.6	12.3	7.0	1.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	10.2	15.8	16.4	21.9	25.4	26.9	25.4	23.9	19.1	13.9	8.9	(27)	
		MCDB	7.3	13.8	21.7	22.1	28.7	30.0	30.5	29.4	28.3	21.1	16.9	10.4	(28)	
	2%	WB	2.4	4.5	13.6	15.2	20.2	23.6	25.5	24.0	22.9	16.4	11.4	5.3	(29)	
		MCDB	3.7	6.0	16.3	20.1	26.2	28.1	28.9	28.2	26.8	20.1	15.0	6.5	(30)	
	5%	WB	1.2	2.2	10.4	13.5	18.7	22.3	24.1	23.0	21.8	14.8	8.8	2.9	(31)	
		MCDB	1.9	3.3	14.4	18.4	23.5	26.1	28.3	26.9	24.6	18.5	12.2	3.9	(32)	
	10%	WB	0.2	0.5	7.7	11.7	17.4	20.9	23.0	22.1	20.5	12.8	7.0	1.4	(33)	
		MCDB	0.6	1.2	10.7	16.0	21.0	25.1	27.2	25.7	23.3	17.0	9.4	2.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	8.9	9.0	10.9	11.6	11.1	11.1	11.5	12.7	11.6	9.9	7.6	(35)	
		MCDBR	9.0	10.2	15.8	16.3	15.8	14.0	12.3	12.8	14.5	16.5	15.0	10.3	(36)	
	5% WB	MCWBR	7.7	8.5	9.3	9.0	6.5	5.8	5.6	6.0	7.0	8.8	9.7	8.3	(37)	
		MCWBR	8.6	9.9	14.7	13.1	11.9	10.5	10.4	10.3	10.8	12.2	14.1	8.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.270	0.288	0.311	0.365	0.394	0.421	0.424	0.408	0.376	0.331	0.290	0.268	(40)	
		taud	2.425	2.382	2.436	2.322	2.309	2.277	2.247	2.290	2.348	2.469	2.510	2.452	(41)	
	Edn at Noon	Edn at Noon	876	915	934	900	880	853	845	847	850	854	847	845	(42)	
		Edh at Noon	74	92	100	122	127	131	134	124	109	85	69	66	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.82	2.87	3.77	4.68	5.50	6.13	6.39	5.40	4.36	2.87	1.91	1.40	(44)	
	RadStd	RadStd	0.19	0.28	0.38	0.47	0.41	0.48	0.31	0.38	0.30	0.37	0.16	0.10	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page